

Homework 9 - Results

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Running Task 1: CPU Scan
CPU-Avg-Time for 100000 elements: 0.23069382 ms
CPU-Avg-Time for 200000 elements: 0.47559738 ms
CPU-Avg-Time for 300000 elements: 0.71887970 ms
CPU-Avg-Time for 400000 elements: 0.96769333 ms
CPU-Avg-Time for 500000 elements: 1.21459961 ms
CPU-Avg-Time for 600000 elements: 1.46558285 ms
CPU-Avg-Time for 700000 elements: 1.85110569 ms
CPU-Avg-Time for 800000 elements: 1.92718506 ms
CPU-Avg-Time for 900000 elements: 2.18701363 ms
CPU-Avg-Time for 1000000 elements: 2.43899822 ms
Running Task 2: GPU Work-Efficient Scan
GPU-Avg-Time for 100000 elements: 0.05141497 ms
GPU-Avg-Time for 200000 elements: 0.05809402 ms
GPU-Avg-Time for 300000 elements: 0.06259298 ms
GPU-Avg-Time for 400000 elements: 0.07170987 ms
GPU-Avg-Time for 500000 elements: 0.07586217 ms
GPU-Avg-Time for 600000 elements: 0.07606101 ms
GPU-Avg-Time for 700000 elements: 0.09311604 ms
GPU-Avg-Time for 800000 elements: 0.08684802 ms
GPU-Avg-Time for 900000 elements: 0.08895993 ms
GPU-Avg-Time for 1000000 elements: 0.09164810 ms
```

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num_elements	CPU	GPU	Speedup-Factor
100000	0.231	0.051	4.53
200000	0.476	0.058	8.21
300000	0.719	0.063	11.41
400000	0.968	0.072	13.44
500000	1.215	0.076	15.99
600000	1.466	0.076	19.29
700000	1.851	0.093	19.91
800000	1.927	0.087	22.15
900000	2.187	0.089	24.57
1000000	2.439	0.092	26.51

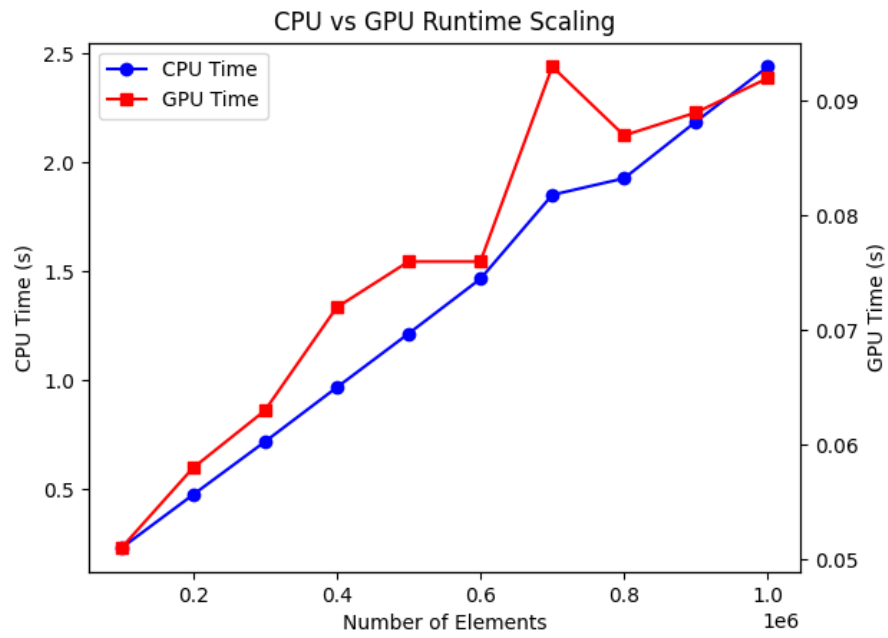


Figure 1: Execution times of the CPU and GPU scan algorithm for different number of elements. (Note: the y-axis are different for CPU and GPU)

Plotted Execution times of GPU and CPU

Plotted Speedup-Factor

Speedup-Factor CPU/GPU of the scan algorithm for different number of elements